

ProShares Replication and Unconstrained Optimization

FINM 36700 Portfolio Theory and Risk Management

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Part I: ProShares Hedge Replication

2 Analyzing the Data

HFRI denotes the HFRI Fund Weighted Composite, MLFM the Merrill Lynch Factor Model (MLEIFCTR), and MLFM-ES its exchange series (MLEIFCTX). All series are monthly and cover Aug 2011 to Aug 2025 (169 months), the period over which HFRI is available.

2.1 Summary statistics

Table 1: Annualized summary statistics of monthly returns

	Mean	Volatility	Sharpe ratio
HFRI	5.1%	5.9%	0.87
MLFM	3.8%	5.5%	0.68
MLFM-ES	3.6%	5.5%	0.65
HDG	2.6%	5.7%	0.45
QAI	2.8%	5.0%	0.56
SPY	14.3%	14.4%	1.00

SPY is shown for reference. Sharpe ratios use total returns.

HFRI has the highest Sharpe ratio of the hedge-fund series (0.87). Its replicators earn less: MLFM 3.8%, HDG 2.6% and QAI 2.8% a year, with similar volatility of about 5 to 6%.

2.2 Tail risk

Table 2: Tail-risk statistics of monthly returns (not annualized)

	Skewness	Excess kurtosis	VaR (.05)	CVaR (.05)	Max drawdown
HFRI	−0.95	5.66	−2.4%	−3.6%	−11.5%
MLFM	−0.28	1.63	−2.7%	−3.5%	−12.4%
MLFM-ES	−0.26	1.59	−2.7%	−3.5%	−12.4%
HDG	−0.26	1.78	−3.0%	−3.7%	−14.1%
QAI	−0.43	1.47	−1.7%	−3.1%	−13.8%
SPY	−0.40	0.69	−6.3%	−8.4%	−23.9%

Table 3: Dates of the maximum drawdown

	Peak	Trough	Recovery
HFRI	Dec 2019	Mar 2020	Aug 2020
MLFM	Jun 2021	Sep 2022	Feb 2024
MLFM-ES	Jun 2021	Sep 2022	Feb 2024
HDG	Jun 2021	Sep 2022	Jul 2024
QAI	Jun 2021	Sep 2022	Feb 2024
SPY	Dec 2021	Sep 2022	Dec 2023

2.3 Regression on SPY

Table 4: Regression of each series on SPY

	Market beta	Treynor ratio	Information ratio
HFRI	0.35	14.8%	0.06
MLFM	0.34	11.0%	−0.45
MLFM-ES	0.34	10.4%	−0.52
HDG	0.35	7.4%	−0.87
QAI	0.30	9.2%	−0.62

Treynor ratio: annualized mean over beta. Information ratio: annualized alpha over annualized residual volatility.

2.4 Discussion

Compared with SPY, the hedge-fund series behave like a scaled-down equity position. SPY earns a much higher mean (14.3%), but with 2.5 to 3 times the volatility, more than twice the VaR and CVaR, and a maximum drawdown of −23.9% against 11–14% for the hedge-fund series. Their market beta is only about 0.35, yet SPY explains 72–79% of their variance, so most of their risk is still market risk.

As for HDG and QAI, we can see that QAI performs better. It has a higher Sharpe ratio (0.56 against 0.45), lower volatility, a smaller VaR (−1.7% against −3.0%), a less negative information ratio and a faster recovery from its drawdown, although both have a negative alpha relative to SPY.

As for whether HDG and the ML series capture the main properties of HFRI, we find that they capture its market exposure but neither its return nor its tails. Their beta (about 0.35), their volatility and their correlation

with HFRI (0.89–0.90) are close to those of HFRI. However, the mean of HFRI is 1.4 to 2.5 percentage points a year higher, and its tails are much fatter, with a skewness of -0.95 and an excess kurtosis of 5.66 against about -0.27 and 1.7 for the replicators. HDG also trails MLFM by about 1.2% a year, which reflects its fees and tracking costs.

2.5 Correlations

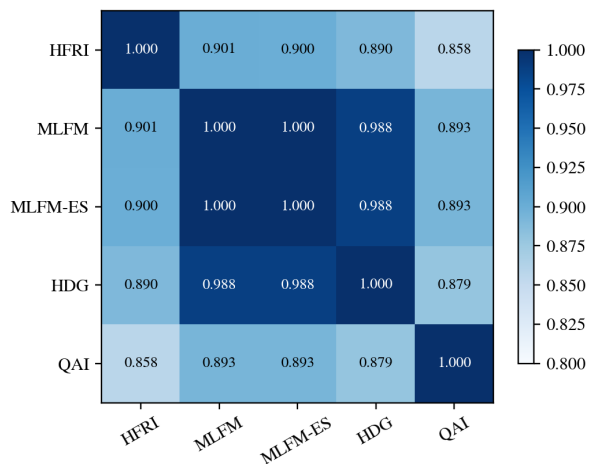


Figure 1: Correlation matrix of monthly returns

The highest correlation is between MLFM and MLFM-ES (0.9999), which are two versions of the same index; HDG, which tracks MLFM-ES, follows at 0.988. The lowest is between HFRI and QAI (0.858).

2.6 Replication of HFRI

The unrestricted regression is

$$r_t^{\text{hfri}} = \alpha + x_t' \beta + \varepsilon_t, \quad \hat{r}_t^{\text{hfri}} = \hat{\alpha} + x_t' \hat{\beta},$$

where x_t holds the six Merrill Lynch factors.

Table 5: Replication of HFRI on the six Merrill Lynch factors

	Estimate
Intercept	0.0011
SPY	0.0435
USGG3M	0.3249
EEM	0.0856
EFA	0.0740
EUO	0.0296
IWM	0.1458
R^2	0.84
Tracking error (annualized)	2.33%

The intercept is monthly.

- (a) The intercept and betas are reported in Table 5. The intercept is 0.0011 a month, about 1.4% a year.
- (b) The betas are realistic position sizes. All six are positive and small, the largest being USGG3M (0.32, a cash-like position) and IWM (0.15). The replication requires no large long-short positions.
- (c) The R^2 is 0.84.
- (d) The tracking error, the annualized volatility of the residual, is 2.33%.

2.7 Out-of-sample replication

For each month from Aug 2016, the regression is estimated on the previous 60 months, and the estimated coefficients are applied to that month's factor returns.

Table 6: Rolling 60-month replication of HFRI, Aug 2016 to Aug 2025

	Out-of-sample	Full-sample OLS, same months
Correlation with HFRI	0.90	0.92
R^2	0.81	0.85
Tracking error (annualized)	2.75%	2.47%
Mean of HFRI (annualized)	6.4%	6.4%
Mean of replication (annualized)	4.8%	5.7%

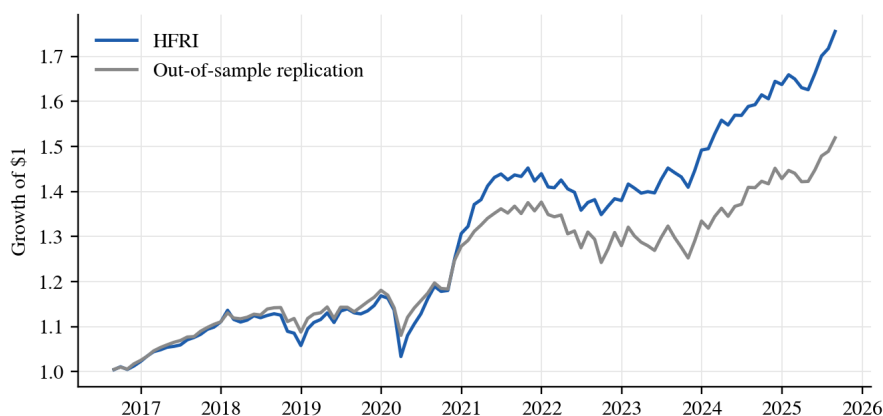


Figure 2: Cumulative returns of HFRI and its out-of-sample replication

The out-of-sample replication tracks HFRI well. Its correlation with HFRI is 0.90 and its R^2 is 0.81, only slightly below the full-sample OLS fit evaluated over the same months (0.92 and 0.85), and its tracking error rises only from 2.47% to 2.75%. The month-to-month movements are therefore captured by stable factor exposures. The level is not: the replication earns 4.8% a year against 6.4% for HFRI, and Figure 2 shows the gap widening over time. The part of HFRI that the factors miss is mostly its average return, the alpha that the replication cannot deliver.

Part II: Unconstrained Optimization

1 Risk Statistics

1.1 Summary statistics

Table 7 reports the requested statistics for each return series.

Table 7: Risk statistics of weekly returns (mean, volatility and Sharpe annualized)

	Mean	Volatility	Sharpe	Skewness	Excess kurtosis	Max drawdown
AAPL	24.9%	27.8%	0.90	−0.18	1.80	−34.6%
NVDA	64.5%	45.4%	1.42	0.35	1.49	−65.9%
MSFT	23.5%	23.8%	0.99	0.13	1.68	−35.1%
GOOGL	27.0%	28.7%	0.94	0.53	3.10	−41.9%
AMZN	26.1%	30.5%	0.86	−0.09	1.34	−54.8%
META	23.2%	35.7%	0.65	0.04	3.49	−76.0%
TSLA	43.7%	58.0%	0.75	0.56	1.63	−72.2%
AVGO	39.4%	38.3%	1.03	0.64	3.27	−40.0%
BRK/B	13.5%	18.8%	0.72	−0.19	2.53	−26.5%
LLY	30.1%	29.8%	1.01	0.14	1.98	−34.5%
SPY	14.5%	16.9%	0.86	−0.58	5.97	−31.8%

Kurtosis is reported in excess of 3. Returns are total returns, as allowed by the assignment.

1.2 Standalone attractiveness

As a standalone investment, NVDA is the most attractive. It has the highest Sharpe ratio (1.42) and by far the highest mean return (64.5% a year), its excess kurtosis is among the lowest, and it has positive skewness (0.35). The cost is high risk: 45.4% volatility and a −65.9% maximum drawdown.

META is the least attractive. It has the lowest Sharpe ratio (0.65), the deepest maximum drawdown (−76.0%), and one of the fattest tails (excess kurtosis 3.49). TSLA is a close second on the downside, with the highest volatility (58.0%) and a −72.2% drawdown.

1.3 Regression on SPY

Table 8: Regression of each stock on SPY, sorted by information ratio

	Alpha	Beta	Information ratio	R^2
NVDA	39.6%	1.72	1.13	0.41
LLY	21.3%	0.61	0.76	0.12
AVGO	18.7%	1.42	0.63	0.40
MSFT	8.8%	1.02	0.53	0.52
GOOGL	11.0%	1.10	0.50	0.42
AAPL	8.6%	1.12	0.42	0.46
AMZN	10.1%	1.11	0.42	0.38
TSLA	18.4%	1.75	0.37	0.26
META	6.0%	1.18	0.20	0.31
BRK/B	2.1%	0.78	0.16	0.49
SPY	0.0%	1.00	–	1.00

Alpha is annualized ($\times 52$); the information ratio is annualized alpha over annualized residual volatility. For SPY on itself the residual is zero, so the information ratio is not defined.

Relative to holding SPY, NVDA is again the most attractive: it has the largest alpha (39.6% a year) and the highest information ratio (1.13). LLY ranks second (information ratio 0.76), with a low beta (0.61) and an R^2 of only 0.12, so most of its risk is unrelated to the market. BRK/B adds the least beyond SPY (information ratio 0.16).

2 Portfolio Allocation

2.1 Correlations

Table 9: Correlation matrix of weekly returns

	AAPL	NVDA	MSFT	GOOGL	AMZN	META	TSLA	AVGO	BRK	LLY	SPY
AAPL	1.00										
NVDA	0.48	1.00									
MSFT	0.55	0.59	1.00								
GOOGL	0.54	0.45	0.61	1.00							
AMZN	0.46	0.55	0.63	0.59	1.00						
META	0.41	0.43	0.57	0.51	0.53	1.00					
TSLA	0.44	0.41	0.41	0.37	0.40	0.26	1.00				
AVGO	0.51	0.57	0.53	0.48	0.44	0.40	0.37	1.00			
BRK	0.41	0.30	0.36	0.34	0.27	0.28	0.20	0.30	1.00		
LLY	0.19	0.16	0.26	0.17	0.14	0.12	0.13	0.10	0.30	1.00	
SPY	0.68	0.64	0.72	0.65	0.61	0.56	0.51	0.63	0.70	0.34	1.00

BRK denotes BRK/B.

The most correlated pair is MSFT and SPY (0.72); among the stocks alone it is MSFT and AMZN (0.63). The least correlated pair is AVGO and LLY (0.10).

LLY should receive extra weight beyond what its mean return alone would justify. It is the least correlated asset in the matrix, with an average correlation of 0.17 with the other stocks and 0.34 with SPY, so it diversifies

the rest of the portfolio.

2.2 Tangency portfolio

Table 10: Tangency portfolio weights and standalone Sharpe ratios

	Tangency weight	Sharpe ratio
BRK/B	1.70	0.72
LLY	0.91	1.01
NVDA	0.85	1.42
GOOGL	0.55	0.94
AVGO	0.45	1.03
AAPL	0.35	0.90
MSFT	0.30	0.99
TSLA	0.20	0.75
AMZN	0.13	0.86
META	0.10	0.65
SPY	−4.54	0.86

Weights sum to one.

The investments with the best Sharpe ratios do not get the biggest weights. BRK/B receives the largest weight (1.70) despite one of the lowest Sharpe ratios (0.72), while SPY, whose Sharpe ratio (0.86) is close to the average, is shorted at −4.54. NVDA, the best Sharpe ratio, comes only third. As expected, LLY gets a large weight (0.91) thanks to its low correlations. The weights depend on the covariance structure, not on the standalone Sharpe ratios.

2.3 Performance of the tangency portfolio

The tangency portfolio has an annualized mean of 102.0%, a volatility of 52.2% and a Sharpe ratio of 1.95.

2.4 Dropping the largest short position

The largest short position is SPY (−4.54). Table 11 compares the weights before and after removing it.

Table 11: Tangency portfolio with and without SPY

	Original	Without SPY
BRK/B	1.70	0.04
LLY	0.91	0.43
NVDA	0.85	0.40
GOOGL	0.55	0.18
AVGO	0.45	0.11
AAPL	0.35	0.01
MSFT	0.30	−0.11
TSLA	0.20	0.02
AMZN	0.13	−0.03
META	0.10	−0.04
SPY	−4.54	
Mean	102.0%	45.4%
Volatility	52.2%	27.2%
Sharpe	1.95	1.67

Mean and volatility are annualized.

Without SPY, the optimized weights are much more reasonable and the gross exposure falls sharply, from 1007.5% to 138.5% (sum of absolute weights). The largest weights now go to LLY (0.43) and NVDA (0.40), the short positions are small, and BRK/B falls from 1.70 to 0.04. The Sharpe ratio drops only from 1.95 to 1.67.

SPY is highly correlated with the individual stocks but has a relatively low mean return (14.5% a year, below every stock except BRK/B), so the unconstrained optimizer uses a very large short position in SPY to hedge a levered long position in the stocks. BRK/B receives a particularly large long weight because it is highly correlated with SPY (0.70) while being much less correlated with the technology stocks (0.27 to 0.41) and having low volatility, which makes it the most effective hedge of the SPY short. This strong collinearity makes the covariance matrix ill-conditioned (its condition number falls from about 207 to 30 once SPY is removed), so the solution is highly sensitive to the estimated means and covariances and produces extreme long-short positions. Once SPY is removed, these hedging opportunities disappear and the weights become much more stable and economically plausible. The small in-sample Sharpe gain of the original portfolio is unlikely to survive out of sample.

A Code

The core computations are reproduced below. The full executed notebook is submitted with this write-up.

Part I: ProShares Hedge Replication

```
import pandas as pd, numpy as np, statsmodels.api as sm

FILE = 'proshares_analysis_data.xlsx'
NAMES = {'HFRI FWI Index': 'HFRI', 'MLEIFCTR Index': 'MLFM', 'MLEIFCTX Index': 'MLFM-ES',
        'HDG US Equity': 'HDG', 'QAI US Equity': 'QAI'}
H = pd.read_excel(FILE, sheet_name='hedge_fund_series', index_col=0).rename(columns=NAMES).dropna()
F = pd.read_excel(FILE, sheet_name='merrill_factors', index_col=0).loc[H.index]
SPY = F['SPY US Equity'].rename('SPY')
P = 12

def summary(R):
    m, v = R.mean() * P, R.std() * np.sqrt(P)
    return pd.DataFrame({'mean': m, 'vol': v, 'sharpe': m / v})

def drawdown(s):
    w = (1 + s).cumprod()
    peak = w.cummax()
    dd = w / peak - 1
    trough = dd.idxmin()
    top = w.loc[:trough].idxmax()
    rec = w.loc[trough:][w.loc[trough:] >= peak.loc[trough]]
    return pd.Series({'max drawdown': dd.min(), 'peak': top, 'trough': trough, 'recovery': rec.index[0] if len(rec) else pd.NaT})

def tail(R):
    q = R.quantile(.05)
    out = pd.DataFrame({'skewness': R.skew(), 'excess kurtosis': R.kurt(), 'VaR (.05)': q, 'CVaR (.05)': R[R <= q].mean()})
    return out.join(pd.DataFrame({c: drawdown(R[c]) for c in R}).T)

def market_reg(R, mkt):
    out = {}
    for c in R:
        fit = sm.OLS(R[c], sm.add_constant(mkt)).fit()
        a, b = fit.params
        out[c] = {'beta': b, 'treynor': R[c].mean() * P / b, 'info ratio': a * P / (fit.resid.std() * np.sqrt(P)), 'r-squared': fit.rsquared}
    return pd.DataFrame(out).T

def replicate(y, X):
    fit = sm.OLS(y, sm.add_constant(X)).fit()
    return fit, pd.Series({'r-squared': fit.rsquared, 'tracking error': fit.resid.std() * np.sqrt(P)})

def rolling_oos(y, X, window=60):
    pred = pd.Series(np.nan, index=y.index)
    for t in range(window, len(y)):
        b = sm.OLS(y.iloc[t - window:t], sm.add_constant(X.iloc[t - window:t])).fit().params
        pred.iloc[t] = b['const'] + X.iloc[t] @ b.drop('const')
```

```

        return pred.dropna()

def oos_stats(y, pred):
    y = y.loc[pred.index]
    e = y - pred
    return pd.Series({'correlation': np.corrcoef(y, pred)[0, 1], 'OOS r-squared': 1 -
        (e ** 2).sum() / ((y - y.mean()) ** 2).sum(),
        'tracking error': e.std() * np.sqrt(P), 'target mean': y.mean() *
        P, 'replication mean': pred.mean() * P})

summary(H.join(SPY))
tail(H.join(SPY))
market_reg(H, SPY)
H.corr()
fit, rep = replicate(H['HFRI'], F)
pred = rolling_oos(H['HFRI'], F)
oos_stats(H['HFRI'], pred)

```

Part II: Unconstrained Optimization

```

import pandas as pd, numpy as np, statsmodels.api as sm

TICKS = ['AAPL', 'NVDA', 'MSFT', 'GOOGL', 'AMZN', 'META', 'TSLA', 'AVGO', 'BRK/B', 'LLY',
        '']
R = pd.read_excel('spx_returns_weekly.xlsx', sheet_name='spx returns', index_col=0)[
    TICKS].join(
    pd.read_excel('spx_returns_weekly.xlsx', sheet_name='additional returns',
        index_col=0)['SPY'])
P = 52

def risk_stats(R):
    m, v = R.mean() * P, R.std() * np.sqrt(P)
    wealth = (1 + R).cumprod()
    return pd.DataFrame({'mean': m, 'vol': v, 'sharpe': m / v, 'skewness': R.skew(),
        'excess kurtosis': R.kurt(), 'max drawdown': (wealth / wealth.
        cummax() - 1).min()})

def regress(y, x):
    fit = sm.OLS(y, sm.add_constant(x)).fit()
    a, b = fit.params
    return pd.Series({'alpha': a * P, 'beta': b, 'info ratio': a * P / (fit.resid.std()
        * np.sqrt(P)), 'r-squared': fit.rsquared})

def optimized_weights(returns, dropna=True, scale_cov=1):
    if dropna:
        returns = returns.dropna()
    covmat_full = returns.cov()
    covmat_diag = np.diag(np.diag(covmat_full))
    covmat = scale_cov * covmat_full + (1 - scale_cov) * covmat_diag
    weights = np.linalg.solve(covmat, returns.mean())
    weights = weights / weights.sum()
    if returns.mean() @ weights < 0:
        weights = -weights
    return pd.DataFrame(weights, index=returns.columns)

def port_stats(w, R):

```

```

p = R @ w
m, v = p.mean() * P, p.std() * np.sqrt(P)
return pd.Series({'mean': m, 'vol': v, 'sharpe': m / v})

S = risk_stats(R)
REG = pd.DataFrame({t: regress(R[t], R['SPY']) for t in TICKS + ['SPY']}).T
REG.loc['SPY', 'info ratio'] = np.nan
C = R.corr()
W = optimized_weights(R)[0]
tangency = port_stats(W, R)
R2 = R.drop(columns=W.idxmin())
W2 = optimized_weights(R2)[0]
tangency_ex = port_stats(W2, R2)
gross = W.abs().sum(), W2.abs().sum()
condition = np.linalg.cond(R.cov()), np.linalg.cond(R2.cov())

```

We used an AI for wording and layout. The analysis and reasoning are our own.